

Concept Question 6.1 Stars and life in the Universe

Imagine that no stars more massive than $2 M_{\odot}$ had ever formed. Would it have been possible for life as we know it to develop? Explain your answer.

Problem 6.2 Galactic chemical evolution

In this exercise, we explore the galactic chemical evolution of the Milky Way using a simplified closed-box model. Our analysis is based on the following assumptions.

- Salpeter's initial mass function, given by $\log N_s = -1.3 \log M + 2.4$, applied within the mass range to be considered, which is between 0.4 to $80 M_{\odot}$.
- The Galaxy begins with an initial gas reservoir of $10^{11} M_{\odot}$ consisting of 76% hydrogen and 24% helium.
- The absolute star formation efficiency remains constant at $4 M_{\odot} \text{yr}^{-1}$.

Let's further assume that in our model the evolution of the chemical composition of the interstellar medium (ISM) is entirely due to the evolution of stars. Specifically, through stellar winds and supernova explosions that take place at the end of a star's life cycle. After the death of a star, the initial stellar mass $M_{\star}$ is split between the mass locked into stellar remnants (white dwarfs and neutron stars) M_{remnant} , the stellar yields – that is the fractions of stellar mass which are transformed into helium M_{He} and heavy elements (C, O, N, ... Fe) M_{Z} –, and the mass of unprocessed material returned to the ISM $M_{\text{unprocessed}}$, so that:

$$\frac{M_{\text{remnant}}}{M_{\star}} + \frac{M_{\text{He}}}{M_{\star}} + \frac{M_{\text{Z}}}{M_{\star}} + \frac{M_{\text{unprocessed}}}{M_{\star}} = f_{\text{remnant}} + f_{\text{He}} + f_{\text{Z}} + f_{\text{unprocessed}} = 1. \quad (1)$$

The mass fractions f_{remnant} , f_{He} , f_{Z} and $f_{\text{unprocessed}}$ depend on initial stellar mass $M_{\star}$ and are given in Table 1 for the considered stellar mass interval. For stellar masses $M_{\star} < 1$ there are no remnant fractions f because these stars have not yet reached the end of their lifetime.

- a) Taking into account main sequence lifetime estimate and plot the number of stars per mass interval $d \log M = 0.1$ after time $t = (10^4, 10^5, 10^6, 10^7, 10^8, 10^9, 10^{10})$ years.
- b) Calculate and plot the time evolution of the total number of stars in the galaxy in the same period of time.
- c) Calculate and plot the evolution of galaxy luminosity in the same period of time.
- d) Derive the mass-to-light ratio for the galaxy in the same period of time.

Table 1: Mass fractions of ejected stellar material for about 24 logarithmic mass bins in the mass interval between 0.4 and $80 M_{\odot}$.

$M_{\star}$	1	1.3	1.6	2	2.5	3.2	4	5	6.3	8
f_{remnant}	0.52	0.42	0.35	0.3	0.25	0.22	0.2	0.16	0.15	0.14
f_{He}	0.02	0.02	0.02	0.03	0.04	0.05	0.07	0.09	0.11	0.12
f_{Z}	0	0	0	0	0	0	0	0	0	0.02
$M_{\star}$	10	12.6	15.8	20	25	32	40	50	63	80
f_{remnant}	0.13	0.12	0.11	0.11	0.11	0.11	0.12	0.13	0.13	0.14
f_{He}	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.1	0.1
f_{Z}	0.05	0.08	0.11	0.15	0.18	0.23	0.25	0.27	0.3	0.32

- e) Calculate and plot the evolution of number of white dwarfs (i.e. all stars with $M < 8 M_{\odot}$ which reached the end of their main sequence lifetime) in a galaxy of age $\log t = (6, 7, 8, 9, 10)$.
- f) Calculate and plot the time evolution of gas mass in the Galaxy as a function of time for $\log t = (4, 5, 6, 7, 8, 9, 10)$.
- g) Calculate and plot the evolution of the total mass of active (burning stars) in the same period of time.
- h) Calculate and plot the evolution of the total mass of stellar remnants as a function of time.
- i) Calculate and plot the evolution of heavy element abundance of the interstellar gas in the same period of time.
- j) Calculate and plot the evolution of helium abundance of the interstellar gas in the same period of time.